STAFF REVIEW · H446 135 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. It references a candidate/primary key in a related table.

2. O(log n) for sorted random-access data.

3. Read-only memory.

4. The longest-waiting item at the front.

I Do — source worked steps

1. Student↔Course is M:N — Find the given data and name the required result.

2. Create Enrolment bridge — Explain why this intermediate stage is necessary.

3. Two one-to-many links — Student(StudentID PK) and Course(CourseID PK) form many-to-many. Resolve with Enrolment(StudentID FK, CourseID FK, EnrolledOn), with composite PK (StudentID, CourseID) if enrolment is unique per pair.

We Do — guided CHECK

1. One Customer
Find the given data and name the required result.
Many Orders
Explain why this intermediate stage is necessary.
FK in Order table
Customer 1:M Order. CustomerID PK in Customer; CustomerID FK in Order. OrderID serves as the Order PK. Optionality (must a customer have an order?) needs a business rule.

You Do — source mark schemes

1. 1:M definition/example [3], M:N definition/example [3].

2. Author(PK) [2], Book(PK) [2], Authorship with two FKs/composite PK [2].

3. Atomicity/1NF problem [2], hard joins/search [2], integrity/update anomaly [2], junction table solution and cardinality [2].

Exit Ticket — source answer

1. False. A junction table is normally required.